

WASP-52b

R-0587

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_230827 · created 2026-07-24T22:06:26+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460183.8798 +/- 0.0076 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.214 +/- 0.063
Transit depth (Rp/Rs)^2	4.58 +/- 2.71 [%]
Orbital Inclination (inc)	89.0 +/- 2.81
Airmass coefficient 1 (a1)	6.48 +/- 3.62
Airmass coefficient 2 (a2)	-1.48 +/- 0.85
Scatter in the residuals of the lightcurve fit is	55.81 %
Best Comparison Star	#3 - [158, 422]
Optimal Aperture	2.72
Optimal Annulus	3.23
Transit Duration (day)	0.0869 +/- 0.0099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.214 ± 0.063 vs 0.1646 (deviation 0.78σ)
Duration fit vs published	0.0869 d vs 0.0758 d (deviation 1.11σ)
Mid-time O–C	-3.8 min
Depth SNR	3.4
Residual scatter	55.81 %

Light curve

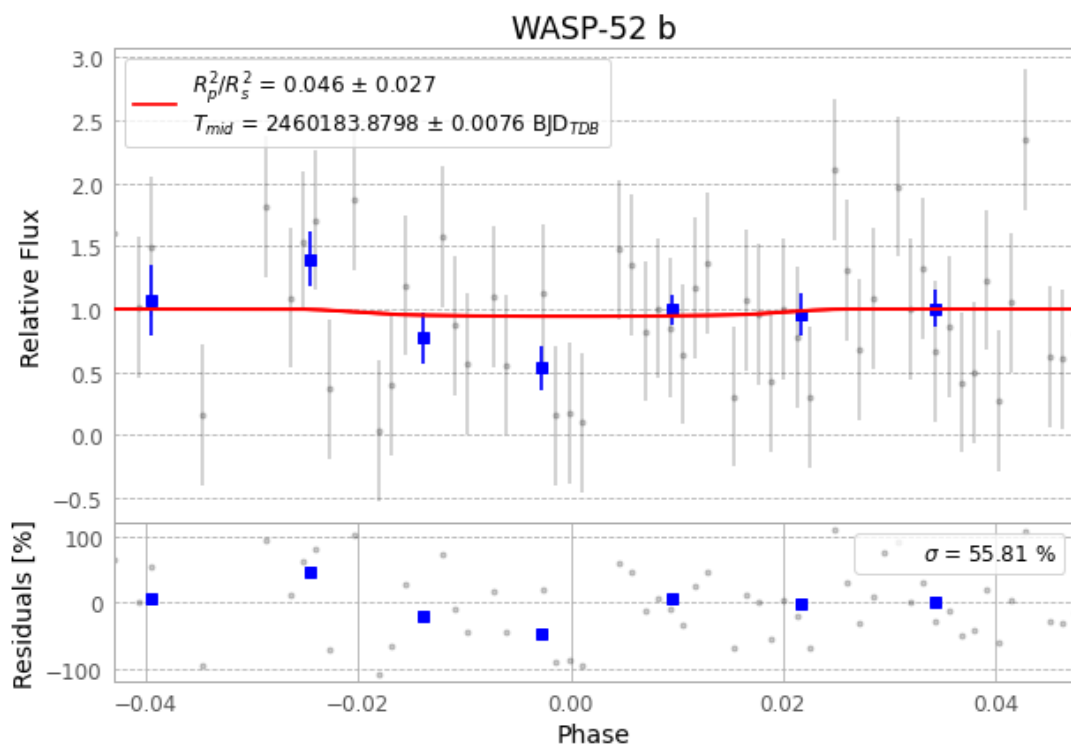


Figure 1 — FinalLightCurve_WASP-52 b_2023-08-27.png (SHA-256 72d373fa0dc9df42...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [552, 423], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 086721259a133ae1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

77 FITS frames (51 MB), WASP-52230827070915.FITS → WASP-52230827105715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230827.log (ff77e51ec428...), FinalLightCurve_WASP-52 b_2023-08-27.png (72d373fa0dc9...), FinalLightCurve_WASP-52 b_2023-08-27.csv (d5c2ee94a129...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 22:06Z.